Chapter 11. Resampling Methods

Contents

1 Training Error and Test Error **1**

2 Cross-Validation **2**

2.1 K-fold cross-validation 2

2.2 Leave-one-out cross-validation 2

2.3 Why cross-validation works: model selection 3

3 The Bootstrap **3**

3.1 A simple example 3

3.2 Back to the real world 4

4 Selected exercises **4**

1 Training Error and Test Error

Resampling methods repeatedly draw samples from the data at hand to learn about an estimator. The two most common are **cross-validation** and the **bootstrap**; they estimate the prediction error, the standard deviation and the bias of estimates.

Definition 11.1

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, g_{D_n} a prediction rule fitted to it, and L a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum (y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation (x_0, y_0) independent of D_n:

$$PE = \mathrm{E}\big[L(y_0, g_{D_n}(x_0))\big].$$

The training error is easy to compute but usually underestimates the test error, because the rule was tuned to the same data: a curve that zig-zags through every training point has zero training error yet predicts new points $y_i^* = f(x_i) + \varepsilon_i^*$ badly. The smaller the test error, the more suitable the model. Measures such as $MS_E = \frac{SS_E}{n-2}$ describe goodness of fit, not predictive ability.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

2 Cross-Validation

2.1 K-fold cross-validation

Randomly divide the data into K roughly equal parts $C_1, \dots, C_K$ (C_k holds the indices of part k, n_k observations; $n_k = n/K$ if n is a multiple of K). For each k, fit the model to the other $K-1$ parts and predict part k:

$$\mathrm{MSE}_k = \frac{1}{n_k} \sum_{i \in C_k} (y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K} \frac{n_k}{n} \mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation i obtained with part k removed.

2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with (x_i, y_i) omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split:

(i) much less bias, since each fit uses $n-1$ observations, almost the whole data set;

(ii) no randomness, since repeated runs give the same result.

Theorem 11.2: LOOCV shortcut for least squares

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{1 - h_{ii}} \right)^2,$$

where $\hat{y}_i$ is the fitted value from the full fit and h_{ii} is the leverage of observation i. The terms $(y_i - \hat{y}_i)/(1 - h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals**; only one fit is needed.

Derivation: proof

Let $\mathbf{x}_i^\top$ be the ith row of $\mathbf{X}$, $A = \mathbf{X}^\top \mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1} \mathbf{x}_i$. Deleting row i gives $A_{(i)} = A - \mathbf{x}_i \mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1} \mathbf{x}_i \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top \mathbf{y}_{(i)} = \mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \left(\mathbf{x}_i^\top A^{-1} + \frac{h_{ii} \mathbf{x}_i^\top A^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1} (\mathbf{X}^\top \mathbf{y} - \mathbf{x}_i y_i)}{1 - h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii} y_i}{1 - h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1 - h_{ii}) y_i - \hat{y}_i + h_{ii} y_i}{1 - h_{ii}} = \dfrac{y_i - \hat{y}_i}{1 - h_{ii}}$.

Remark 11.3: Choice of K

LOOCV does not shake up the data enough: the n fits are trained on nearly identical sets, so their errors are highly correlated and their average can have high variance. $K = 5$ or 10 is usually a better choice.

2.3 Why cross-validation works: model selection

Example 11.4: Choosing the degree of a polynomial

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select p. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big(f(x_i) - \hat{f}_p(x_i)\big)^2,$$

but f is unknown. Consider instead the prediction squared error for new responses $y_i^* = f(x_i) + \varepsilon_i^*$:

$$\begin{aligned} PSE(p) &= \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big(y_i^* - \hat{f}_p(x_i)\big)^2 = \frac{1}{n}\sum_{i=1}^{n} \mathrm{E}\big[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*\big]^2 \\ &= \mathrm{MSE}(p) + \sigma^2, \end{aligned}$$

because ε_i^* is independent of $\hat{f}_p$ and has mean zero. So minimizing PSE is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n} \big(y_i - \hat{f}_p^{-i}(x_i)\big)^2,$$

where $\hat{f}_p^{-i}$ is fitted without the ith point. Since y_i is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}\big(y_i - \hat{f}_p^{-i}(x_i)\big)^2 = \sigma^2 + \mathrm{E}\big(f(x_i) - \hat{f}_p^{-i}(x_i)\big)^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$ (leaving out one point changes little), $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$. Thus $\hat{p} = \arg\min_p CV(p)$ is a sensible rule. The training error $\frac{1}{n}\sum(y_i - \hat{f}_p(x_i))^2$ does not have this property: it keeps decreasing as p grows.

3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

3.1 A simple example

We invest a fraction α of our money in an asset with return X and $1-\alpha$ in one with return Y, and choose α to minimize the risk $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem 11.5

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X - Y) \geq 0$, this is a minimum. ■

The variances are unknown, so we plug in sample estimates to get $\hat{\alpha}$. To learn the standard deviation of $\hat{\alpha}$ in a simulation, we could draw 1,000 independent data sets of 100 pairs from the population, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000} \hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000}(\hat{\alpha}_r - \bar{\alpha})^2}.$$

3.2 Back to the real world

For real data we cannot draw new samples from the population. The bootstrap instead draws data sets *from the original data set with replacement*, using the computer to mimic the process of generating new samples.

Definition 11.6: Bootstrap standard error

1. From the original data Z of size n, draw a bootstrap data set Z^{*1} of size n by sampling observations with replacement (some observations appear more than once, some not at all).
2. Compute the estimate $\hat{\alpha}^{*1}$ from Z^{*1}.
3. Repeat B times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}$.
4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\alpha}^*\right)^2}, \qquad \bar{\alpha}^* = \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}.$$

Example 11.7: Three observations

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\ldots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.

4 Selected exercises

Motivation. The first item, from the resampling chapter of *An Introduction to Statistical Learning*, explains the familiar "63%" of a bootstrap sample. The others make three claims of the chapter

exact: what the bootstrap standard error converges to, why LOOCV estimates the prediction error, and how much the training error is too optimistic.

Exercise 11.1 (Which observations enter a bootstrap sample)

A bootstrap sample of size n is drawn with replacement from n observations. (After ISLR, Ch. 5, Exercise 2.)
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that the jth observation is not in the bootstrap sample is $(1-1/n)^n$, and find its limit as $n \to \infty$.
(c) Find the probability that the jth observation is in the bootstrap sample for $n = 5, 100, 10{,}000$, and the expected number of distinct original observations in a bootstrap sample.

Solution.

(a) Each bootstrap draw selects one of the n observations with probability $1/n$, so it is not observation j with probability $1-1/n$. The second draw is made from the same full set, independently of the first, so the answer is again $1-1/n$.

(b)

Step 1. The n draws are independent, so $P(j \notin Z^*) = \prod_{r=1}^{n} P(\text{draw } r \neq j) = (1-1/n)^n$.

Step 2. $\log(1-1/n)^n = n\log(1-1/n) = -1 - \frac{1}{2n} - \frac{1}{3n^2} - \cdots \to -1$, using $\log(1-u) = -u - u^2/2 - \cdots$ for $|u| < 1$. Hence $(1-1/n)^n \to e^{-1} = 0.368$.

(c)

Step 1. $P(j \in Z^*) = 1-(1-1/n)^n$ equals $1-0.8^5 = 0.672$ for $n=5$, $1-0.99^{100} = 0.634$ for $n=100$ and 0.632 for $n=10{,}000$, close to the limit $1-e^{-1} = 0.632$.

Step 2. The number of distinct observations is $\sum_{j=1}^{n} \mathbf{1}\{j \in Z^*\}$, so by linearity its mean is $n\left[1-(1-1/n)^n\right] \approx 0.632\, n$.

About 37% of the observations are left out of each bootstrap sample; these "out-of-bag" observations can serve as a test set for the rule fitted to that sample. ■

Exercise 11.2 (Ideal bootstrap standard error of the mean)

Let $x_1, \ldots, x_n$ be fixed data and $\bar{x}^*$ the mean of a bootstrap sample $x_1^*, \ldots, x_n^*$. Writing $\mathrm{E}^*, \mathrm{Var}^*$ for moments under bootstrap resampling, show that

$$\mathrm{E}^*(\bar{x}^*) = \bar{x}, \qquad \mathrm{Var}^*(\bar{x}^*) = \frac{\hat{\sigma}^2}{n}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2,$$

and conclude that $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ as $B \to \infty$.

Solution. [Supplement]

Step 1. The x_r^* are iid, each equal to x_i with probability $1/n$. Hence $\mathrm{E}^* x_r^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^* x_r^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$.

Step 2. By independence, $\mathrm{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\bar{x}^* = \hat{\sigma}^2/n$.

Step 3. The bootstrap replicates $\bar{x}^{*1}, \ldots, \bar{x}^{*B}$ are iid draws from this resampling distribution, so by the law of large numbers their sample variance converges to $\mathrm{Var}^*(\bar{x}^*)$, and $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ almost surely.

The limit is the plug-in standard error $\sqrt{(n-1)/n}\,s/\sqrt{n}$, slightly below the usual $s/\sqrt{n}$: the bootstrap treats the data as the population, with divisor n. The Monte Carlo error from finite B is a separate, controllable source of error. ■

Exercise 11.3 (LOOCV for the sample mean)

Predict every response by the sample mean, $\hat{f} \equiv \bar{y}$.
(a) Show that $y_i - \bar{y}_{(i)} = \frac{n}{n-1}(y_i - \bar{y})$, where $\bar{y}_{(i)}$ is the mean without observation i, and hence $CV_{(n)} = \left(\frac{n}{n-1}\right)^2 \frac{1}{n}\sum_i (y_i - \bar{y})^2$. Check this against the LOOCV shortcut.
(b) If $y_1, \dots, y_n$ are iid with mean μ and variance σ^2, show that $\mathrm{E}\, CV_{(n)} = \sigma^2(1 + \frac{1}{n-1})$, the exact prediction error of a mean computed from $n-1$ observations.

Solution. [Supplement]

(a) $\bar{y}_{(i)} = \frac{n\bar{y} - y_i}{n-1}$, so $y_i - \bar{y}_{(i)} = \frac{(n-1)y_i - n\bar{y} + y_i}{n-1} = \frac{n(y_i - \bar{y})}{n-1}$. Squaring and averaging gives $CV_{(n)}$. The mean is LS regression on the constant column $\mathbf{X} = \mathbf{1}$, with $\mathbf{H} = \frac{1}{n}\mathbf{J}$ and $h_{ii} = \frac{1}{n}$; the shortcut gives $(y_i - \bar{y})/(1 - \frac{1}{n})$, the same residual.

(b)

Step 1. $\mathrm{E}\sum_i (y_i - \bar{y})^2 = (n-1)\sigma^2$ for iid data (the unbiasedness of s^2). Hence $\mathrm{E}\, CV_{(n)} = \frac{n^2}{(n-1)^2} \cdot \frac{(n-1)\sigma^2}{n} = \frac{n\sigma^2}{n-1} = \sigma^2(1 + \frac{1}{n-1})$.

Step 2. For a new y_0 independent of a sample of size $n-1$ with mean $\bar{y}_{n-1}$, $\mathrm{E}(y_0 - \bar{y}_{n-1})^2 = \mathrm{Var} y_0 + \mathrm{Var}\bar{y}_{n-1} = \sigma^2 + \frac{\sigma^2}{n-1}$, the same value.

So LOOCV is exactly unbiased for the prediction error of the rule fitted to $n-1$ points, and slightly pessimistic for the rule fitted to all n (whose error is $\sigma^2(1 + \frac{1}{n})$), as the polynomial example suggested. ■

Exercise 11.4 (Optimism of the training error)

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ matrix of rank p, $\mathrm{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}\boldsymbol{\varepsilon} = \sigma^2\mathbf{I}$, and $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ the LS fit. Let $\mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*$ be new responses at the same x's, with $\boldsymbol{\varepsilon}^*$ independent of $\boldsymbol{\varepsilon}$ and distributed like it. Show that

$$\mathrm{E}\Big[\frac{1}{n}\|\mathbf{y} - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 - \frac{p}{n}\Big), \qquad \mathrm{E}\Big[\frac{1}{n}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 + \frac{p}{n}\Big).$$

Solution. [Supplement]

Step 1. (Facts on $\mathbf{H}$.) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\big[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}\mathbf{I}_p = p$. For a random vector $\mathbf{u}$ with mean $\mathbf{0}$ and $\mathrm{Var}\mathbf{u} = \sigma^2\mathbf{I}$ and a fixed matrix A, $\mathrm{E}(\mathbf{u}^\top A\mathbf{u}) = \mathrm{tr}(A\,\mathrm{E}\mathbf{u}\mathbf{u}^\top) = \sigma^2\mathrm{tr}A$.

Step 2. (Training error.) $(\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, so $\mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ and $\|\mathbf{y} - \hat{\mathbf{y}}\|^2 = \boldsymbol{\varepsilon}^\top(\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$. By Step 1 its mean is $\sigma^2(n-p)$.

Step 3. (Prediction error.) $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\varepsilon}$, so $\mathbf{y}^* - \hat{\mathbf{y}} = \boldsymbol{\varepsilon}^* - \mathbf{H}\boldsymbol{\varepsilon}$ and

$$\mathrm{E}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2 = \mathrm{E}\|\boldsymbol{\varepsilon}^*\|^2 - 2\,\mathrm{E}(\boldsymbol{\varepsilon}^{*\top}\mathbf{H}\boldsymbol{\varepsilon}) + \mathrm{E}(\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}) = n\sigma^2 - 0 + p\sigma^2,$$

the middle term vanishing by independence and $\mathrm{E}\boldsymbol{\varepsilon}^* = \mathbf{0}$.

Step 4. Dividing by n gives both formulas. The gap $2p\sigma^2/n$ grows with the number of parameters: the training error decreases as the model grows while the prediction error eventually increases, which is why model selection uses cross-validation (or a penalty such as $2p\hat{\sigma}^2/n$) rather than the training error. ■